

2016.0 RANGE ROVER (LG), 501-14

HANDLES, LOCKS, LATCHES AND ENTRY SYSTEMS

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the Locks, Latches and Entry Systems, refer to the relevant Description and Operation section in the workshop manual.

REFER to: [Handles, Locks, Latches and Entry Systems](#) (501-14 Handles, Locks, Latches and Entry Systems, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Doors, hood and tailgate alignment ▪ Fuel filler door actuator ▪ Hood release handle ▪ Hood release cables ▪ Hood latches ▪ Door latches and strikers ▪ Tailgate latch and striker ▪ Door exterior handles ▪ Door interior handles ▪ Tailgate release switch	▪ Fuses ▪ Wiring harnesses and connectors ▪ Central junction box ▪ Keyless vehicle module ▪ Smart keys

3. If an obvious cause for an observed or reported concern is found,

correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index
5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

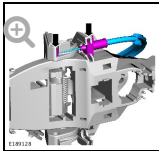
SYMPTOM	POSSIBLE CAUSES	ACTION
Front or rear door will not open when using the exterior door handle - Interior door handle functions normally	▪ Exterior door handle damaged ▪ Exterior door handle release cable disconnected from the exterior door handle ▪ Exterior door handle release cable disconnected from the door latch ▪ Exterior door handle release cable twisted ▪ Door latch internal failure	▪ GO to Pinpoint Test A.
Front or rear door will not open when using the interior door handle - Exterior door handle functions normally	▪ Child locks engaged - Rear doors only ▪ Interior door handle damaged ▪ Interior door handle release	▪ GO to Pinpoint Test B.

	cable disconnected from the interior door handle ▪ Interior door handle release cable disconnected from the door latch ▪ Interior door handle release cable retaining clip and/or housing damaged ▪ Door latch internal failure	
NOTE: The vehicle will not lock if the hood, the tailgate or a door is detected as open. The battery back-up sounder or passive sounder will emit a double chirp to signify a mis-lock. Message center indicates that a door is open when it is physically closed	▪ Door striker adjustment incorrect ▪ Door ajar switch circuit open circuit ▪ Door latch internal failure - Door ajar switch	▪ GO to Pinpoint Test C.
Door will not lock/unlock - One door only	▪ Central door locking motor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Door latch internal failure	▪ GO to Pinpoint Test H.
Doors will not lock/unlock - All doors	▪ Central door locking system	▪ Using the Jaguar Land Rover approved diagnostic

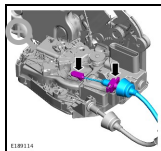
	fault	equipment, check the central junction box for related DTCs and refer to the relevant DTC index
Door will not lock/unlock mechanically using the emergency key barrel	▪ Exterior door handle damaged ▪ Emergency key barrel cable disconnected from the emergency key barrel ▪ Emergency key barrel cable disconnected from the door latch ▪ Door latch internal failure	▪ GO to Pinpoint Test E.
Fuel flap will not lock or unlock	▪ Fuel flap release motor detached from mounting bracket ▪ Fuel flap release motor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel flap release motor internal failure	▪ Check the integrity of the fuel flap release motor installation ▪ Refer to the electrical circuit diagrams and check the fuel flap release motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the central junction box for related DTCs and refer to the relevant DTC index
Soft close inoperative or rattles	▪ Closing motor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Closing motor cable disconnected	▪ GO to Pinpoint Test I.

	from the door latch ▪ Door striker maladjusted ▪ Closing motor internal failure ▪ Door latch internal failure	
Door will not close fully	▪ Door latch spring installed incorrectly ▪ Door latch internal failure	▪ GO to Pinpoint Test G.

PINPOINT TESTS

PINPOINT TEST A : EXTERIOR DOOR HANDLE TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: EXTERIOR DOOR HANDLE TEST 1	
	1 Remove the interior door trim as necessary. REFER to: (501-05 Interior Trim and Ornamentation) Front Door Trim Panel (Removal and Installation), Rear Door Trim Panel - Standard Wheel Base (Removal and Installation).
	2 Check the exterior door handle release cable for correct installation to the exterior door handle.
	Is the exterior door handle release cable correctly installed to the exterior door handle? Yes GO to A2. No Install the exterior door handle release cable to the exterior door handle. If the exterior door handle release cable is damaged, install a new exterior door handle release cable. GO to A4.

A2: EXTERIOR DOOR HANDLE TEST 2



- 1 Check the exterior door handle release cable for correct installation to the door latch.

Is the exterior door handle release cable correctly installed to the door latch?

Yes

GO to A3.

No

Install the exterior door handle release cable to the door latch. If the exterior door handle release cable is damaged, install a new exterior door handle release cable. **GO to A4.**

A3: EXTERIOR DOOR HANDLE TEST 3

- 1 Check the exterior door handle inner release cable for twists.

Is the exterior door handle inner release cable twisted?

Yes

Disconnect the exterior door handle release cable from the latch. Allow the exterior door handle inner release cable to untwist. Re-connect the exterior door handle release cable. **GO to A4.**

No

GO to A4.

A4: EXTERIOR DOOR HANDLE TEST 4

- 1 Open the door.

- 2 Using a suitable tool, close the door latch claw to the fully closed position.

- 3 Using the exterior door handle, attempt to open the door latch.

Did the door latch open?

Yes

GO to A5.

No

Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems)

Front Door Latch (Removal and Installation),

Rear Door Latch - Standard Wheel Base (Removal and Installation).

A5: EXTERIOR DOOR HANDLE TEST 5

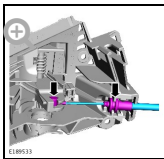
- 1 Check the operation of the exterior door handle and door latch:
 - Close the door fully.
 - Using the smart key, lock the vehicle.

	NOTES: ▪ The door should not open during this sub-step. ▪ The smart key must be out of range of the vehicle during this sub-step - Passive entry vehicles only. ▪ Using the exterior door handle, attempt to open the door. ▪ Using the smart key, unlock the vehicle. NOTE: The door should open during this sub-step. ▪ Using the exterior door handle, attempt to open the door.
	2 Perform step 1 a total of five times.
	Did the exterior door handle and door latch function correctly during every test? Yes Repair complete. No Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation).

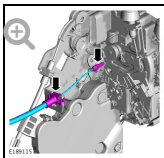
NOTE:

If the door is a rear door, ensure that the child locks are not engaged and the fault is still present before continuing.

PINPOINT TEST B : INTERIOR DOOR HANDLE TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: INTERIOR DOOR HANDLE TEST 1	
	1 Remove the interior door trim as necessary. REFER to: (501-05 Interior Trim and Ornamentation)

	Front Door Trim Panel (Removal and Installation), Rear Door Trim Panel - Standard Wheel Base (Removal and Installation).
	2 Check the interior door handle release cable for correct installation to the interior door handle.
	Is the interior door handle release cable correctly installed to the interior door handle? Yes GO to B2. No Install the interior door handle release cable to the interior door handle. If the interior door handle release cable is damaged, install a new interior door handle release cable. GO to B4.

B2: INTERIOR DOOR HANDLE TEST 2

	NOTE: For vehicles without soft closing, there is a redundant cable bracket on the door latch housing. The interior door handle release cable must not be installed to this cable bracket. 1 Check the interior door handle release cable for correct installation to the door latch.
	Is the interior door handle release cable correctly installed to the door latch? Yes GO to B3. No Install the interior door handle release cable to the door latch. If the interior door handle release cable is damaged, install a new interior door handle release cable. GO to B4.

B3: INTERIOR DOOR HANDLE TEST 3

	1 Check the interior door handle release cable retaining clip and housing at the door latch for damage.
	Is the retaining clip and/or housing damaged? Yes Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation). No

	GO to B4.
B4: INTERIOR DOOR HANDLE TEST 4	
	1 Open the door.
	2 Using a suitable tool, close the door latch claw to the fully latched position.
	3 Using the interior door handle, attempt to open the door latch.
	Did the door latch open? Yes GO to B5. No Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation).
B5: INTERIOR DOOR HANDLE TEST 5	
	1 Check the operation of the interior door handle and door latch: ▪ Close the door fully. ▪ Using the smart key, double lock the vehicle. NOTE: The door should not open during this sub-step. ▪ Using the interior door handle, attempt to open the door. ▪ Using the smart key, unlock the vehicle. NOTE: The door should open during this sub-step. ▪ Using the interior door handle, attempt to open the door.
	2 Perform step 1 a total of five times.
	Did the interior door handle and door latch function correctly during every test? Yes Repair complete. No

Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems)

[Front Door Latch](#) (Removal and Installation),

[Rear Door Latch - Standard Wheel Base](#) (Removal and Installation).

NOTE:

Ensure that the door latch claw remains in the fully latched position during this test.

NOTE:

Ensure that the door latch claw remains in the fully latched position during this test.

PINPOINT TEST C : DOOR AJAR TESTS

**TEST
CONDITIONS**

DETAILS/RESULTS/ACTIONS

C1: DOOR AJAR TEST 1

1 Open the door.

2 Using a suitable tool, close the door latch claw to the fully latched position.

3 Set the ignition to on.

4 Check the message center for a 'DOOR OPEN' message.

Does the message center indicate that the door is open?

Yes

GO to C2.

No

Adjust the door striker outwards by 1 mm. Check that panel flushness has not been affected by this adjustment.

REFER to: [Body and Frame](#) (501-26 Body Repairs - Vehicle Specific Information and Tolerance Checks, Description and Operation).

Retest.

C2: DOOR AJAR TEST 2

1 Remove the interior door trim as necessary. REFER to: (501-05 Interior Trim and Ornamentation)

	Front Door Trim Panel (Removal and Installation), Rear Door Trim Panel - Standard Wheel Base (Removal and Installation).
	2 Check the door latch electrical connector for correct installation.
	Is the door latch electrical connector installed correctly? Yes GO to C3. No Install the door latch electrical connector correctly and retest.
C3: DOOR AJAR TEST 3	
	1 Disconnect the door latch electrical connector.
	2 Check the door latch electrical connector terminals for damage and corrosion.
	Are the door latch electrical connector terminals damaged/corroded? Yes Repair the door latch electrical connector or install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation). No GO to C4.
C4: DOOR AJAR TEST 4	
	1 Re-connect the door latch electrical connector.
	2 Set the ignition to on.
	3 Check the message center for a 'DOOR OPEN' message.
	Does the message center indicate that the door is open? Yes GO to C5. No GO to C6.
C5: DOOR AJAR TEST 5	
	1 Disconnect the door latch electrical connector.
	NOTE: This step applies when testing the front left or rear left door.
	2 Using a suitable link wire, temporarily connect terminal 3 of the wiring harness connector to terminal 4 of the wiring harness connector.

	NOTE: This step applies when testing the front right or rear right door. 3 Using a suitable link wire, temporarily connect terminal 6 of the wiring harness connector to terminal 5 of the wiring harness connector.
	4 Check the message center for a 'DOOR OPEN' message.
	Does the message center indicate that the door is open? Yes Refer to the electrical circuit diagrams and check the door ajar switch circuit for open circuit, high resistance. Repair the wiring harness as necessary. No Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation).
C6: DOOR AJAR TEST 6	
	1 Re-connect the door latch electrical connector.
	2 Check the operation of the door ajar switch: - Close the door. NOTE: A DOOR OPEN message should not be displayed. - Check the message center for a DOOR OPEN message. - Open the door. NOTE: A DOOR OPEN message should be displayed. - Check the message center for a DOOR OPEN message.
	3 Perform step 2 a total of five times.
	Did the door ajar switch function correctly? Yes

	Repair complete. No Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation).
--	---

PINPOINT TEST D : DOOR LOCKING TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: DOOR LOCKING TEST 1	
	1 Remove the interior door trim as necessary. REFER to: (501-05 Interior Trim and Ornamentation) Front Door Trim Panel (Removal and Installation), Rear Door Trim Panel - Standard Wheel Base (Removal and Installation).
	2 Check the door latch electrical connector for correct installation.
	Is the door latch electrical connector installed correctly? Yes GO to D2. No Install the door latch electrical connector correctly and retest.
D2: DOOR LOCKING TEST 2	
	1 Disconnect the door latch electrical connector.
	2 Check the door latch electrical connector terminals for damage and corrosion.
	Are the door latch electrical connector terminals damaged/corroded? Yes Repair the door latch electrical connector or install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation). No GO to D3.
D3: DOOR LOCKING TEST 3	
	NOTE: This step applies when testing the front left or rear left door. 1 Connect a voltmeter to terminal 6 and terminal 7 of the door latch

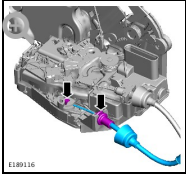
	wiring harness connector.
	NOTE: This step applies when testing the front right or rear right door. 2 Connect a voltmeter to terminal 3 and terminal 2 of the door latch wiring harness connector.
	3 Using the smart key, lock the vehicle and check the measured voltage.
	Is the measured voltage greater than 10.0 V? Yes GO to D4. No Refer to the electrical circuit diagrams and check the door latch single locking circuit for short circuit to ground, short circuit to power, open circuit, high resistance.

D4: DOOR LOCKING TEST 4

	1 Using the smart key, unlock the vehicle.
	NOTE: This step applies when testing the front left or rear left door. 2 Connect a voltmeter to terminal 8 and terminal 7 of the door latch wiring harness connector.
	NOTE: This step applies when testing the front right or rear right door. 3 Connect a voltmeter to terminal 1 and terminal 2 of the door latch wiring harness connector.
	4 Using the smart key, double lock the vehicle and check the measured voltage.
	Is the measured voltage greater than 10.0 V? Yes Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation). No

Refer to the electrical circuit diagrams and check the door latch double locking circuit for short circuit to ground, short circuit to power, open circuit, high resistance.

PINPOINT TEST E : EMERGENCY KEY BARREL TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
E1: EMERGENCY KEY BARREL TEST 1	
	1 Remove the interior door trim. REFER to: Front Door Trim Panel (501-05 Interior Trim and Ornamentation, Removal and Installation).
	2 Check the emergency key barrel cable for correct installation to the door latch.
	Is the emergency key barrel cable correctly installed to the door latch? Yes GO to E2. No Install the emergency key barrel cable to the door latch. If the emergency key barrel cable is damaged, install a new emergency key barrel cable. GO to E4.
E2: EMERGENCY KEY BARREL TEST 2	
	1 Using a suitable tool, close the door latch claw to the fully latched position.
	2 Using the emergency key, lock and unlock the door. Check that the emergency key barrel cable pushes and pulls the door latch linkage.
	Does the door latch linkage move when the door is locked and unlocked? Yes GO to E3. No Install the emergency key barrel cable to the exterior door handle. If the emergency key barrel cable is damaged, install a new emergency key barrel cable. GO to E4.
E3: EMERGENCY KEY BARREL TEST 3	
	1 Open the door.

	2 Using a suitable tool, close the door latch claw to the fully latched position.
	3 Using the emergency key barrel, lock the door.
	4 Using the exterior door handle, attempt to open the door latch.
	Did the door latch open? Yes Install a new door latch. REFER to: Front Door Latch (501-14 Handles, Locks, Latches and Entry Systems, Removal and Installation). No GO to E4.

E4: EMERGENCY KEY BARREL TEST 4

	1 Check the operation of the emergency key barrel and door latch: - Close the door fully. - Using the emergency key barrel, lock the door. NOTES: - The door should not open during this sub-step. - The smart key must be out of range of the vehicle during this sub-step - Passive entry vehicles only. - Using the exterior door handle, attempt to open the door. - Using the emergency key barrel, unlock the door. NOTE: The door should open during this sub-step. - Using the exterior door handle, attempt to open the door.
	2 Perform step 1 a total of five times.
	Did the emergency key barrel and door latch function correctly during every test? Yes Repair complete. No Install a new door latch. REFER to: Front Door Latch (501-14 Handles, Locks, Latches and Entry Systems, Removal and Installation).

NOTE:

This test measures the resistance of the soft close initiation switch.

NOTE:

This test measures the resistance of the soft close initiation switch.

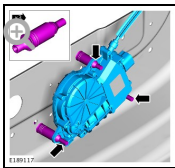
NOTE:

This test measures the resistance of the soft close position switch.

NOTE:

This test measures the resistance of the soft close position switch.

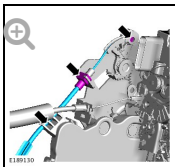
PINPOINT TEST F : SOFT CLOSE TESTS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
F1: SOFT CLOSE TEST 1	
	1 Open the door.
	2 Using a suitable tool, close the door latch claw to the first position and then gently apply pressure to the claw to open the door latch.
	Did the soft close feature operate and close the door latch fully? Yes GO to F5. No GO to F2.
F2: SOFT CLOSE TEST 2	
	1 Remove the interior door trim as necessary. REFER to: (501-05 Interior Trim and Ornamentation)

	Front Door Trim Panel (Removal and Installation), Rear Door Trim Panel - Standard Wheel Base (Removal and Installation).
	2 Check the closing motor for correct installation.
	3 Check the closing motor electrical connector for correct installation.
	Are the closing motor and electrical connector correctly installed? Yes GO to F3. No Install the closing motor correctly. If the closing motor is damaged, install a new closing motor. Retest.

F3: SOFT CLOSE TEST 3

	1 Disconnect the closing motor electrical connector.
	2 Check the closing motor electrical connector terminals for damage and corrosion.
	Are the closing motor electrical connector terminals damaged/corroded? Yes Repair the closing motor electrical connector or install a new closing motor as necessary. No GO to F4.

F4: SOFT CLOSE TEST 4

	1 Check the closing motor cable for correct installation to the door latch.
	Is the closing motor cable correctly installed to the door latch? Yes GO to F5. No Install the closing motor cable to the door latch. If the closing motor cable is damaged, install a new closing motor and cable.

F5: SOFT CLOSE TEST 5

	1 Using the exterior door handle, open the door latch.
--	---

	2 Close the door to the first position of the door latch.
	WARNING: Do not place fingers in the door aperture as the soft close feature may operate and close the door fully. Failure to follow this instruction may result in personal injury. NOTE: Open the window during this step to provide access. 3 Without operating the exterior door handle, gently pull the door outwards.
	Did the soft close feature operate and close the door latch fully? Yes Adjust the door striker inwards by 1 mm. Check that panel flushness has not been affected by this adjustment. REFER to: Body and Frame (501-26 Body Repairs - Vehicle Specific Information and Tolerance Checks, Description and Operation). Retest. No GO to F6.

F6: SOFT CLOSE TEST 6

	1 Using the exterior door handle, open the door latch.
	2 Remove the door latch electrical connector.
	NOTES: ▪ This step applies when testing the front left or rear left door. ▪ Circuit reference - PAWL / GND - 3 Connect an ohm-meter to terminal 3 and terminal 5 of the door latch.

	NOTES: ▪ This step applies when testing the front right or rear right door. ▪ Circuit reference - PAWL / GND - 4 Connect an ohm-meter to terminal 4 and terminal 2 of the door latch.
	Is the measured resistance greater than 10.0 MΩ? Yes GO to F7. No Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation).

F7: SOFT CLOSE TEST 7

	1 Using a suitable tool, close the door latch claw to the first position.
	NOTES: ▪ This step applies when testing the front left or rear left door. ▪ Circuit reference - PAWL / GND - 2 Connect an ohm-meter to terminal 3 and terminal 5 of the door latch.
	NOTES: ▪ This step applies when testing the front right or rear right door. ▪ Circuit reference - PAWL / GND - 3 Connect an ohm-meter to terminal 4 and terminal 2 of the door latch.
	Is the measured resistance less than 10.0 Ω? Yes GO to F8. No Install a new door latch as necessary. REFER to: (501-14 Handles,

Locks, Latches and Entry Systems)
[Front Door Latch](#) (Removal and Installation),
[Rear Door Latch - Standard Wheel Base](#) (Removal and Installation).

F8: SOFT CLOSE TEST 8

1 Using the exterior door handle, open the door latch.

NOTES:

- This step applies when testing the front left or rear left door.
- Circuit reference - CLAW / GND -

2 Connect an ohm-meter to terminal 4 and terminal 5 of the door latch.

NOTES:

- This step applies when testing the front right or rear right door.
- Circuit reference - CLAW / GND -

3 Connect an ohm-meter to terminal 3 and terminal 2 of the door latch.

Is the measured resistance less than 10.0 Ω ?

Yes

GO to F9.

No

Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems)

[Front Door Latch](#) (Removal and Installation),

[Rear Door Latch - Standard Wheel Base](#) (Removal and Installation).

F9: SOFT CLOSE TEST 9

1 Using a suitable tool, close the door latch claw to the first position.

NOTES:

- This step applies when testing the front left or rear left door.
- Circuit reference - CLAW / GND -

	2 Connect an ohm-meter to terminal 4 and terminal 5 of the door latch.
	NOTES: ▪ This step applies when testing the front right or rear right door. ▪ Circuit reference - CLAW / GND - 3 Connect an ohm-meter to terminal 3 and terminal 2 of the door latch.
	Is the measured resistance less than 10.0 Ω? Yes GO to F10. No Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation).

F10: SOFT CLOSE TEST 10

	NOTE: Circuit reference - PAWL - 1 Refer to the electrical circuit diagrams and check the soft close initiation switch signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance.
	Is there a wiring harness fault? Yes Repair the wiring harness as necessary. No GO to F11.

F11: SOFT CLOSE TEST 11

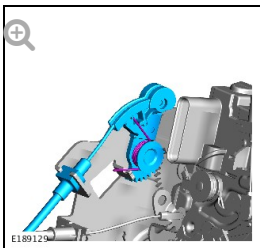
	NOTE: Circuit reference - CLAW - 1 Refer to the electrical circuit diagrams and check the soft close position switch signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance.
--	---

	Is there a wiring harness fault? Yes Repair the wiring harness as necessary. No GO to F12.
--	--

F12: SOFT CLOSE TEST 12

	1 Refer to the electrical circuit diagrams and check the closing motor power and ground circuits for short circuit to ground, short circuit to power, open circuit, high resistance.
	Is there a wiring harness fault? Yes Repair the wiring harness as necessary. No Install a new closing motor.

PINPOINT TEST G : DOOR NOT LATCHING TESTS

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
G1: DOOR NOT LATCHING TEST 1	
	1 Remove the interior door trim as necessary. REFER to: (501-05 Interior Trim and Ornamentation) Front Door Trim Panel (Removal and Installation), Rear Door Trim Panel - Standard Wheel Base (Removal and Installation).
	2 Remove the door latch.
	3 Check that the spring is installed correctly.
	Is the spring installed correctly? Yes Install a new closing motor and door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation). No Install the spring correctly. Re-install the door latch. GO to G2.
G2: DOOR NOT LATCHING TEST 2	

	1 Check the operation of the door latch: - Close the door fully. - Using the smart key, lock the door. NOTES: - The door should not open during this sub-step. - The smart key must be out of range of the vehicle during this sub-step - Passive entry vehicles only. - Using the exterior door handle, attempt to open the door. - Using the smart key, unlock the door. NOTE: The door should open during this sub-step. - Using the exterior door handle, attempt to open the door.
	2 Perform step 1 a total of five times.
	Did the door latch function correctly during every test? Yes Repair complete. No Install a new door latch as necessary. REFER to: (501-14 Handles, Locks, Latches and Entry Systems) Front Door Latch (Removal and Installation), Rear Door Latch - Standard Wheel Base (Removal and Installation).

PINPOINT TEST H : NON - INTRUSIVE TEST - DOOR LATCH	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
H1: NON - INTRUSIVE TEST - DOOR LATCH	
	1 Using the Jaguar Land Rover approved diagnostic equipment, connect to the vehicle.
	2 Select the appropriate diagnosis symptoms: Body – Security and Locking – Handles Locks and Latches

	3 Under the recommendations section select the "Run" button next the following application: ▪ Inline diagnostic unit 2 diagnostic test - Door latch (Unilatch)
	4 Following the on screen instructions. You will be guided to run a test from the fuse of the suspect door without removing any trims or panels.
	Did the component pass the test? Yes No further action needed. Make sure the correct suspect door has been interrogated. If the problem persistsGO to Pinpoint Test D . No GO to Pinpoint Test D .

PINPOINT TEST I : NON - INTRUSIVE TEST - CLOSING MOTOR	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
I1: NON - INTRUSIVE TEST - CLOSING MOTOR	
	1 Using the manufacturer approved diagnostic system connect to the vehicle.
	2 Select the appropriate diagnosis symptoms: ▪ Body – Security and Locking – Handles Locks and Latches
	3 Under the recommendations section select the "Run" button next the following application: ▪ Inline diagnostic unit 2 diagnostic test - Closing motor
	4 Following the on screen instructions. You will be guided to run a test from the fuse of the suspect door without removing any trims or panels.
	Did the component pass the test? Yes No further action needed. The part performed inside the normal operation conditions. No GO to Pinpoint Test F .

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this

vehicle, please refer to Section 100-00. REFER to: (100-00 General Information)

[Diagnostic Trouble Code Index - DTC: Driver/Passenger Door Module \(DDM/PDM\)](#) (Description and Operation),

[Diagnostic Trouble Code Index - DTC: Body Control Module \(BCM\)](#) (Description and Operation),

[Diagnostic Trouble Code Index - DTC: Keyless Vehicle Module \(KVM\)](#) (Description and Operation),

[Diagnostic Trouble Code Index - DTC: Tailgate Control Module \(TGCM\) - Upper/Lower](#) (Description and Operation).